Critical power test for ramp exercise

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LETTER TO THE EDITOR

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Sirs:

Hugh Morton has recently proposed a critical power test for ramp exercise on a cycle ergometer (Morton 1994). This new test was assumed to be more reliable than the usual critical power test the protocol of which consists in performing several periods of constant-load exercise up to exhaustion. Indeed, it was postulated that the reliability of exhaustion time of long lasting exercise was low in comparison with the reliability of the duration of a ramp test which was assumed to depend less on motivation. According to Morton's model, the duration (T) of ramp test depends on critical power (CP), ramp slope (S) and anaerobic work capacity (AWC), only:

$$T = CP/S + 2\ AWC \quad (1)$$

However, the validity of this procedure is questionable. Although T depends on CP and AWC, maximal aerobic power (P_{max}) is probably the main physiological factor determining ramp exercise duration. In fact, the model proposed by Morton implicitly assumed that CP was equal to P_{max}. Indeed, as shown in Fig. 1 of Morton's paper, all the energy was assumed to be supplied by the anaerobic metabolism beyond CP and consequently CP corresponded to P_{max}. Morton's paper should have been entitled "Maximal aerobic power test for ramp exercise test".

However, anaerobic metabolism is involved in energy supply during submaximal exercise and CP has been shown to be clearly lower than P_{max} in all the studies on this topic (Moritani et al. 1981; Poole et al. 1988; Vandewalle et al. 1989; McLellan and Cheung 1992).

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Is it possible to apply the method proposed by Morton to the calculation of the physiological factors limiting the duration of a ramp test? In theory, ramp test duration should depend on S, P_{max}, AWC and the contribution of anaerobic metabolism to energy supply at submaximal exercise. If P_{anaer} is the power output corresponding to the anaerobic metabolism at a given power P, then we can write for a ramp test:

$$P_{anaer} = f(P) = f(S \cdot t) \quad (2)$$

where t = time

$$W_{anaer} = \int_0^T f(P)\,dt = \int_0^T f(S \cdot t)\,dt = AWC \quad (3)$$

where W_{anaer} is the work produced by anaerobic metabolism.

If P_{anaer} is negligible for P lower than the power output corresponding to the current concept of anaerobic threshold (A) we can write:

$$AWC = \int_A^T f(P)\,dt = \int_A^T f(S \cdot t)\,dt \quad (4)$$
$$= f(S, T, A)$$

In this case, whatever the relationship between P and P_{anaer}, P_{anaer} should be equal to zero for $P = A$, and dP_{anaer}/dP should be asymptotic to 1 for P close to P_{max}. A complementary exponential or a hyperbolic function can be proposed as examples of this kind of relationship between P_{anaer} and P for P greater than A.

$$P_{anaer} = X - (1 - e^{-kX})/k$$

$$P_{anaer} = X - 1/k + 1/(X + k)$$

where k is a constant and $X = P - A$

Unfortunately, after integration of these equations, the relationship between T and S is not explicit, i.e. the dependent variable T cannot be expressed directly in terms of the independent variable $S (T = f(S))$. Consequently, it is not possible to estimate AWC, A, and

P_{max} by means of the fitting procedure for nonlinear equations provided in standard statistical software, as suggested by Morton. However, given the speed of available microcomputers, it should be possible to design a programme dedicated to the calculation of these parameters according to the least square method and giving results within a few minutes. Nevertheless, it is likely that the reliability and accuracy of the estimations of P_{max}, A and AWC by this method would be largely limited by the sensitivity of such models to small variations in T.

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